

Chapter 50. Pain Control, Sedation, and Neuromuscular Blockade

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Figure 50-1.

Pain Control, Sedation, and Neuromuscular Blockade: Introduction

Mechanically ventilated patients often require sedatives and analgesics, and in some circumstances may even require the addition of neuromuscular blockade. The myriad disease processes causing respiratory failure frequently elicit a sense of respiratory distress in these patients. In addition, therapies such as endotracheal intubation and positive-pressure ventilation bring about discomfort to a significant number of patients in the intensive care unit (ICU), and patients often receive these life support treatments after surgical interventions or medical conditions that themselves carry a burden of pain. Accordingly, most patients receive analgesics and/or sedatives while undergoing mechanical ventilation. Although many drugs are available to carry out the goals of pain control and, if necessary, sedation in the ICU, studies of the use of these agents were initially performed in other settings such as the operating room or procedure suite. However, accumulation of evidence directed at ICU patient outcomes—specifically patients undergoing mechanical ventilation—has increased over recent decades, and awareness of the complex pharmacology of sedatives and analgesics in critical care is now well established. At the same time, heightened awareness of the complications associated with the use of neuromuscular blocking agents has relegated their use to mostly short-term indications, such as general anesthesia in the operating room, with ICU use now reserved for a small population of patients, typically suffering from severe acute respiratory distress syndrome. The drugs used to achieve analgesia and sedation of mechanically ventilated patients are necessarily potent; however, the enduring effects when these drugs are used without discretion has impacted strategies for their administration.

Rationale

Recognition of the fundamental pathophysiologies of respiratory failure and the interaction of the mechanical ventilator and patient will allow goals of pain control, sedation, and neuromuscular blockade to complement the supportive goals of mechanical ventilation. An individualized strategy can then be executed, recognizing goals unique to each patient. This chapter discusses principles and goals of pain control, sedation, and neuromuscular blockade in mechanically ventilated patients, as well as review the drugs currently available to achieve these goals.

Pain Control

We begin our discussion with the analgesic agents because many patients experience pain to some degree, because inadequate control of pain is unfortunately a memory patients may have after ICU management, and because the ability of the patient to clearly indicate and quantify pain may be significantly impaired by respiratory failure and the devices employed to treat it. This last fact can lead to the unfortunate circumstance of pain in the mechanically ventilated patient presenting as agitation, which, if treated with sedatives alone, is often inadequately managed and leads to excessive drug administration.

Indications

The indications for analgesia during mechanical ventilation come directly from the multiple reasons for pain in these patients. Pain from surgical incisions or trauma is self-evident, but other indications for pain control are subtler. These include endotracheal suctioning or placement of invasive catheters such as arterial or venous lines. Preexisting problems such as skeletal fractures from metastatic cancer

and prolonged immobility during bed rest¹ should be considered as other potential causes of pain. The mere presence of an endotracheal tube causes pain in some patients.

Pain may cause many adverse effects, including increased endogenous catecholamine activity, myocardial ischemia, hypercoagulability, hypermetabolic states, sleep deprivation, anxiety, and delirium.² Treating pain diminishes some of these detrimental effects.³

Assessing Adequacy

It is undeniable that pain occurs in some mechanically ventilated patients.⁴⁻⁶ Failure to recognize that pain can cause agitation that may lead to inappropriate administration of nonanalgesic sedatives. Accordingly, an aggressive approach to managing pain has been strongly recommended by consensus opinions regarding sedation of mechanically ventilated patients.^{7,8} Indeed, it is prudent to actively inquire for the presence of pain and begin the assessment of mechanically ventilated patients by postulating that pain is present as a symptom. The ability to accurately discern pain may be difficult because many ventilated patients are unable to convey its presence and clinical parameters such as changes in vital signs are often not reliable indicators.

In spite of such recommendations and vigilance, the management of pain in ventilated patients is inadequate in some cases.⁵ Ineffective communication with patients may be at the root of this problem, because delirium in the ICU is common.⁹ Concern over addiction to opiates,¹⁰ adverse cardiopulmonary effects of analgesics, and arbitrary dose limits are other reasons for inadequate analgesia in ventilated patients.

Although attention to analgesia is important, it is similarly important to recognize that not all ventilated patients experience pain. A recent prospective observational study of 171 ICU patients described their self-reported experiences while in the ICU (34% of these were mechanically ventilated).⁴ Only 40% of patients reported pain. The important message from this study is that pain, although common and distressing, is not universal. Accordingly, a concerted effort to determine whether or not a patient has pain is extremely important, lest unnecessary withholding or administration of analgesics occur.

Tools to categorize pain, such as scales or scoring systems, may be beneficial. In general, simpler scales are more effective because communication for many ventilated patients is limited. The visual analog scale, although not specifically evaluated in critically ill, mechanically ventilated patients, has very good reliability and validity.^{11,12} This scale uses a self-reported measure of pain intensity that consists of a 10-cm line on paper with verbal anchors of “no pain” and “severe pain.” A similar scale is the numeric rating scale. This scale also consists of a horizontal line with numeric markings 1 and 10 anchoring either extreme of the pain-intensity scale.¹³

Previous studies have shown that benzodiazepines may enhance the analgesic effects of opiates^{14,15} and that opiate requirements are decreased in patients sedated with benzodiazepines compared with propofol.^{16,17} Notwithstanding this interesting observation, it is imperative that sedative agents are not used in the place of analgesics.

Selection of Agent

Nonpharmacologic analgesic strategies are worth considering. For example, malpositioning of invasive catheters (an endotracheal tube impinging on the main carina) is a problem that may be easily remedied. Likewise, optimal patient positioning in bed may at least partially relieve low back pain, pain from chest tubes, and so on. In spite of appropriate attention to nonpharmacologic approaches, most patients require administration of pharmacologic agents. Opiates are given in most circumstances and are discussed below.

Opiates

Opiate receptors are ubiquitous, being found in the central nervous system as well as peripheral tissues. The two most clinically important opiate receptors are μ and κ . There are two subtypes of μ receptors, μ_1 and μ_2 . Of these subtypes μ_1 receptors mediate analgesia, while μ_2 receptors mediate respiratory depression, nausea, vomiting, constipation, and euphoria. The κ receptors are responsible for such effects as sedation, miosis, and spinal analgesia. Several opiate drugs are available for clinical use and are discussed in detail below.

Morphine

Morphine is a commonly used opiate agent and is the drug with which all other opiates are typically compared. Administered intravenously, it has a relatively slow onset of action (typically 5 to 10 minutes) because of its low lipid solubility, which delays movement of the drug across the blood–brain barrier. The duration of action after a single intravenous dose of **morphine** is approximately 4 hours. When given repeatedly, accumulation in tissue stores may prolong the effect. It is metabolized in the liver, undergoing glucuronide conjugation, and its active metabolite, morphine-6-glucuronide, may accumulate, particularly in renal failure. Elimination occurs in the kidney, so effects may be prolonged in renal failure. The primary central effect of all opiates is analgesia, which is mediated through the μ and κ receptors. Opiates have some anxiolytic properties, but no reliable amnesia.

Fentanyl

Fentanyl is a synthetic opiate. It has a very high lipid solubility, unlike **morphine**, leading to rapid movement across the blood–brain barrier and rapid onset of action. Redistribution into peripheral tissues leads to short duration of action after a single dose (0.5 to 1 hour). When fentanyl is given repeatedly and tissue stores become saturated, its clinical effect can be prolonged. Fentanyl has no active metabolites and does not release histamine.¹⁸

Hydromorphone

Hydromorphone has pharmacologic properties similar to **morphine** with regard to onset of action. Because it has no active metabolites, however, the duration of action after long periods of hydromorphone administration may be less than that of **morphine**.

Meperidine

Meperidine is a lipid-soluble opiate with rapid movement across the blood–brain barrier and a rapid onset of action (3 to 5 minutes). The higher lipid solubility leads to peripheral redistribution so that duration after a single dose may be shorter as compared with **morphine** (1 to 4 hours). Meperidine undergoes hepatic metabolism and renal elimination. Because metabolites of meperidine may accumulate secondary to the common organ dysfunctions that accrue in critically ill patients, and because these metabolites have neurotoxic effects that may cause seizures, there seems no reason to ever use this agent in the ICU patient (see below).¹⁹

Remifentanyl

Remifentanyl is another synthetic, lipid-soluble drug with a rapid onset of action. It is quickly metabolized via hydrolysis by nonspecific blood and tissue esterases. As such, its pharmacokinetic profile is not affected by hepatic or renal insufficiency. It is given by continuous infusion because of its rapid metabolism and recovery time. When remifentanyl is used during general anesthesia, available data suggest that postoperative respiratory failure is reduced, presumably because patients wake up more rapidly. It provides reliable short-term analgesia in patients with postoperative respiratory failure,²⁰ and may reduce ICU admissions by allowing extubation in the operating room.²¹ Because of remifentanyl's unique pharmacology, there is a potential for rapid tolerance to the drug.²² This has been referred to as *hyperalgesia*.²³ Because remifentanyl is eliminated from the body so rapidly and completely, it may lead to a circumstance where patients are left with no analgesia. Although further research is needed to more fully address this issue, it is important to acknowledge this potential problem when using remifentanyl in the ICU patient.

Methods of Delivery

For most patients, intravenous injection of opiate analgesics is the preferred route. Intramuscular injections are discouraged because of pain related to the injection itself and unpredictable absorption in critically ill patients. Dosing strategies for intravenous administration of opiates include continuous infusions and intermittent dosing strategies. Intermittent dosing can be further divided into scheduled administration, administration on an *as needed* or *PRN* basis, and patient-controlled analgesia. Dosing given as needed may lead to fluctuations between inadequate and excessive analgesia. Patients alert enough to respond to their own pain needs may benefit from patient-controlled analgesia strategies,²⁴ although most ventilated patients are not alert enough to utilize a patient-controlled analgesia device. Transdermal opiates may be continued in patients who are chronically receiving such medications, although transcutaneous

absorption is unpredictable during critical illness. Certainly, this route should not be used for treating acute pain in mechanically ventilated patients.

Toxicity

All opiates induce respiratory depression, which is centrally mediated and dose dependent. Depression is mediated by the μ_2 receptors in the brainstem medulla; the typical pattern is one of reduced respiratory rate but preserved tidal volume. The response to hypercapnia is decreased and the ventilatory response to hypoxia is obliterated. The depressed respiratory property of these drugs can be exploited in ventilated patients suffering from subjective dyspnea or coughing.

The hemodynamic effects of opiates in euvoletic patients are typically minimal. Hypovolemic patients with blood pressure sustained by sympathetic hyperactivity may suffer hypotension after the administration of opiates. Most opiates cause a decrease in heart rate because of decreased sympathetic activity. Although [morphine](#) causes histamine release, this does not usually cause hemodynamic compromise. Remifentanyl may cause bradycardia and hypotension, particularly when administered concurrently with drugs having vasodilating properties, such as propofol. Hypertension after remifentanyl, though described, is uncommon.

Gastrointestinal dysfunction in the form of pharmacologic ileus is common with the use of opiates in mechanically ventilated patients. Methylnaltrexone, a specific antagonist of μ_2 receptors in the gut, has been reported to attenuate opiate-induced gastrointestinal ileus in humans.^{25,26} The utility of methylnaltrexone in critically ill, mechanically ventilated patients has not been tested.

Seizures are a toxicity unique to meperidine and occur as a result of its metabolite normeperidine, which is a central nervous system stimulant. Patients with renal failure and/or with prolonged use are especially prone to normeperidine accumulation. Because meperidine offers no significant advantage over other available opiates, its side-effect profile should largely preclude its use in critically ill patients.

Muscle rigidity occasionally occurs with synthetic opiates such as fentanyl and remifentanyl. It is not seen with naturally occurring opiates like [morphine](#). It is usually seen when high doses of these drugs are injected rapidly and may affect chest wall muscles. The mechanism of opiate-induced skeletal muscle rigidity, although not fully understood, is thought to involve supraspinal activity of the drugs in the striata and substantia nigra. In the most extreme cases, respiratory muscle rigidity makes ventilation impossible. Neuromuscular blockade, typically with succinylcholine, reverses this problem. Fortunately, this problem is extremely rare with the doses of opiates used in the management of ventilated patients in the ICU.

Dependence and withdrawal can be seen in patients receiving opiates for extended time periods when the drugs are discontinued. Patients who abuse opiates are at risk for this when hospitalized during critical illness. The signs and symptoms seen in withdrawal syndromes are mostly nonspecific and include pupillary dilation, sweating, lacrimation, rhinorrhea, piloerection, tachycardia, vomiting, diarrhea, hypertension, yawning, fever, tachypnea, restlessness, irritability, increased sensitivity to pain, nausea, cramps, muscle aches, dysphoria, insomnia, symptoms of opioid craving, and anxiety. Patients without previous illicit drug use may also experience opiate withdrawal when pharmacologically administered opiates given for extended time periods are suddenly stopped. Whether any preemptive strategies, such as downward titration or regular interruption of dosing, will attenuate or prevent opiate withdrawal is not known. Although apparently logical, use of long-acting opiates (e.g., methadone, transdermal fentanyl) to overcome this problem has not been studied in the ICU. [Table 50-1](#) summarizes the pharmacologic properties of commonly used opiates.

Table 50-1: Commonly Used Opiates

	Morphine	Fentanyl	Remifentanyl	Methadone
Typical starting dose	2 to 5 mg	25 to 50 mcg	0.025 to 0.1 mcg/kg/min infusion	5 to 10 mg
Onset	10 minutes	0.5 to 1 minute	3 to 5 minutes	10 to 20 minutes
Duration after single dose	4 hours	0.5 to 1 hour	NA; 5 minutes after 4-hour infusion	6 to 24 hours
Metabolism	Hepatic	Hepatic	Nonspecific tissue and plasma esterases	Hepatic
Elimination	Renal	Renal	Renal	Renal
Anxiolysis	+	++	++	+
Analgesia	++++	++++	++++	++++
Hypnosis	No reliable effect	No reliable effect	No reliable effect	No reliable effect
Amnesia	No reliable effect	No reliable effect	No reliable effect	No reliable effect
Seizure threshold	No effect	No effect	No effect	No effect
Reducing dyspnea	++++	++++	++++	++++
Cardiovascular effect	Venodilation	Venodilation	Venodilation	Venodilation
Respiratory effect	Hypoventilation	Hypoventilation	Hypoventilation	Hypoventilation
Common side effects	Ileus, nausea and vomiting, itching	Ileus, nausea and vomiting, itching	Ileus, nausea and vomiting, itching	Ileus, nausea and vomiting, itching

+, minimal effect; ++, mild effect; +++, moderate effect; +++++, large effect.

Sedation

Indications

Sedation needs vary widely in mechanically ventilated patients. Although nonpharmacologic approaches such as comfortable positioning in bed and verbal reassurance are reasonable initial considerations, treatment with sedative agents is frequently needed. Effective use of sedatives and analgesics in critically ill patients begins with an understanding of their various indications. Effective *analgesia* (discussed above) is extremely important and should be considered concurrently with sedation.

Anxiety occurs frequently in ventilated patients and is one of the most common indications for sedation. Anxiety may result from (a) uncertainty of one's surroundings, diagnosis, or prognosis; (b) uncomfortable experiences such as the presence of an endotracheal tube

or invasive diagnostic or therapeutic procedures; or (c) isolation from family and friends. Sedatives may be required to facilitate routine nursing care of ventilated patients. Procedures such as endotracheal suctioning, dressing changes, and repositioning regularly elicit distress requiring sedatives.

Dyspnea is common in ventilated patients and may be a source of distress requiring sedation. Excessive coughing may contribute to patient-ventilator dyssynchrony in some patients. Many patients requiring mechanical ventilation suffer from cardiopulmonary instability and impaired gas exchange. Abnormal elevations in [oxygen](#) consumption and carbon dioxide production may compromise such patients. Reduction of [oxygen](#) consumption can stabilize the balance between [oxygen](#) supply and demand.¹⁶ Autonomic instability and elevated endogenous catecholamine activity are common in ventilated patients and may lead to hemodynamic changes (e.g., tachycardia or hypertension) that may elicit myocardial ischemia in some patients, particularly those at risk for coronary artery disease.²⁷

Amnesia is often cited as an indication for sedation of ventilated patients. Certainly, for those patients mechanically ventilated during surgical procedures, the importance of amnesia is indisputable.²⁸ On the contrary, in ICU patients, the necessity of continuous amnesia is far less certain. Although amnesia for certain portions of critical illness requiring mechanical ventilation may seem logical (e.g., during invasive procedures), complete amnesia for extended periods during mechanical ventilation in the ICU has never been proven to confer benefit. Indeed, some data suggest that prolonged ICU amnesia may be detrimental to long-term neuropsychiatric recovery from critical illness.^{29–31} It is clear, however, that complete amnesia is mandatory whenever neuromuscular blocking agents are administered.

The relationship between *delirium* and sedation is indisputable. Delirium—defined as an acutely changing or fluctuating mental state, inattention, disorganized thinking, and an altered level of consciousness that may or may not be accompanied by agitation—has been noted to have a wide incidence range in critically ill patients.^{9,32,33} Hyperactive or agitated delirium may occur with anxiety as well as sepsis, fevers, encephalopathy (e.g., hepatic or renal), withdrawal syndromes ([alcohol](#), tobacco, or other illicit drugs), or medications. Such hyperactive delirium is frequently treated with neuroleptic medications such as haloperidol.³⁴ A substantial fraction, however, of ventilated ICU patients exhibit an insidious, hypoactive form of delirium. Currently, it is unclear whether this form of delirium is an indication for any pharmacologic therapy.

Assessing Adequacy

The assessment of sedation adequacy requires a thorough awareness of the indications for sedation. Even when indications are clear, adequacy assessment is challenging because of its subjective nature. As such, a reliable instrument to categorize level of sedation is an important initial step. In view of that, several sedation scales such as the Ramsay Sedation Score,³⁵ the Sedation Agitation Scale,³⁶ and the Richmond Agitation-Sedation Scale (RASS)³⁷ have been developed. The Ramsay scale, described nearly 40 years ago, is often referenced in clinical investigations of sedation, even though it has never been objectively validated.³⁸ Accordingly, other sedation scales such as the Sedation Agitation Scale and RASS have been developed and tested for validity and reliability.³⁹ The RASS is perhaps the most extensively evaluated scale. It has been validated for ability to detect changes in sedation status over consecutive days of ICU care, as well as against constructs of level of consciousness and delirium. Because of their more precise scientific evaluations, the RASS and Sedation Agitation Scale should be viewed as preferable to the traditional Ramsay scale.

Ultimately, the evaluation of sedation adequacy is an individualized bedside maneuver. Guidance from nurses is useful, because they are often the first to notice changes in level of sedation. Theoretically, the optimal level of sedation would result in a state in which all indications for sedation are attended to, and the patient is fully communicative with bedside caregivers. This state of being awake and communicative while sedatives are being administered is ideal, and is possible in many patients. Unfortunately, for some, the stresses of respiratory failure and mechanical ventilation do not permit such a state.

Objective monitors of sedation level would be attractive tools for ventilated patients. The bispectral index monitor processes raw electroencephalogram signals into a discrete scaled number from 0 (absence of cortical activity) to 100 (fully awake). This monitor has been shown to track the level of consciousness under general anesthesia. There are some data that suggest a good correlation between the bispectral index and the sedation agitation scale⁴⁰ as well as the RASS,³⁹ however, problems such as electromyographic interference⁴¹ have undermined the utility of this device for widespread use in the ICU.

Selection of Agent

After indications for sedation are established, the clinician must choose one or more agents to administer. This section outlines the various agents currently available and discusses the pharmacologic properties of each agent.

Benzodiazepines

Benzodiazepines act by potentiating γ -aminobutyric acid (GABA) receptor-mediated inhibition of the central nervous system. The GABA receptor complex regulates a chloride channel on the cell membrane, and, by increasing the intracellular flow of chloride ions, neurons become hyperpolarized, with a higher threshold for excitability. The GABA receptor can be competitively antagonized with the synthetic agent flumazenil, thereby reversing the pharmacologic effects of benzodiazepines. Midazolam, lorazepam, and diazepam are the three available parenteral benzodiazepines.

The onset of action of midazolam is rapid (0.5 to 5 minutes) and the duration following a single dose is short (approximately 2 hours). All parenteral benzodiazepines, including midazolam, are lipid soluble with a large volume of distribution, and are therefore widely distributed throughout body tissues. For all benzodiazepines, the duration of action after a single bolus depends mainly on the rate of redistribution to peripheral tissues, especially adipose tissue. Midazolam undergoes hepatic metabolism and renal excretion. One metabolite of midazolam (α -hydroxymidazolam) has active pharmacologic properties and is renally excreted. Accordingly, there seems little point to using midazolam in ICU patients who have kidney dysfunction. In the presence of normal renal function, α -hydroxymidazolam has a half-life of 1 hour. The pharmacokinetics of midazolam are very different when it is administered to critically ill patients, especially when given by continuous infusion for prolonged periods. Under these circumstances there is preferential accumulation of the drug in peripheral tissues, where metabolism is not possible. When drug infusions are stopped, the peripheral tissue stores release midazolam back into the bloodstream, leading to prolongation of clinical effect.⁴² Obese and elderly patients may be even more prone to prolonged effects.

Intravenous lorazepam has a slower onset of action when compared to midazolam (5 minutes). This is the result of a lower lipid solubility that slows the drug's ability to cross the blood–brain barrier. The duration of action following a single dose is long (6 to 10 hours) and is proportional to the dose given. Because these data refer to healthy volunteers, applicability to critically ill patients is difficult. Lorazepam's longer duration of action is caused by lower lipid solubility with decreased peripheral tissue redistribution. The absence of renally excreted metabolites makes it the benzodiazepine of choice in patients with kidney failure.

Intravenous diazepam has a rapid onset of action (1 to 3 minutes) and a limited duration of action following a single dose (30 to 60 minutes). Like midazolam, these pharmacologic properties are caused by the high lipid solubility, which leads to rapid peripheral redistribution of diazepam and extended duration of action. As such, there is little point to ever using diazepam for ICU sedation.

The parenteral benzodiazepines have similar pharmacodynamic effects. All cause dose-dependent suppression of awareness along a spectrum from mild depression of responsiveness to obtundation. They are potent anxiolytic drugs and reliably produce amnesia;⁴³ lorazepam appears to produce the longest duration of antegrade amnesia. Another common feature of these drugs is their reliable anticonvulsant effect.⁴⁴ Rarely, benzodiazepines cause agitation. This paradoxical response typically accelerates as additional benzodiazepine is given; although rare, it has a propensity to affect elderly patients. All benzodiazepines induce dose-dependent depression of respiratory drive. This ventilatory depression, while less extreme than that seen with opiates, behaves in a synergistic manner with opiate-induced respiratory depression. A pattern of reduced tidal volume and slightly increased respiratory rate may help to discern a benzodiazepine effect from the pattern of slow, deep breathing typically seen with opiates. Benzodiazepines, like opiates, can obliterate the hypoxic ventilatory drive. They have minimal effects on the cardiovascular system of euvolemic patients, typically causing slight decreases in blood pressure without a significant change in heart rate. The effect is exacerbated in the presence of relative hypovolemia. In patients with elevated endogenous sympathetic drive, more profound decreases in blood pressure can be seen. [Table 50-2](#) summarizes the pharmacologic properties of the parenteral benzodiazepines.

Table 50-2: Properties of Commonly Used Benzodiazepines

	Midazolam	Lorazepam	Diazepam
Typical starting dose	1 to 2 mg	0.5 to 1 mg	5 to 10 mg
Onset	0.5 to 2 minutes	3 to 5 minutes	1 to 3 minutes
Duration after single dose	2 hours	6 to 10 hours	1 to 6 hours
Metabolism	Hepatic	Hepatic (less influenced by age and liver disease)	Hepatic
Elimination	Renal	Renal	Renal
Anxiolysis	++++	++++	++++
Analgesia	No effect	No effect	No effect
Hypnosis	++++	++++	++++
Amnesia	++++	++++	++++
Seizure threshold	+++	++++	+++
Reducing dyspnea	+	+	+
Cardiovascular effect	Venodilation	Venodilation	Venodilation
Respiratory effect	Hypoventilation	Hypoventilation	Hypoventilation
Common side effects	Paradoxical agitation	Paradoxical agitation	Paradoxical agitation

+, minimal effect; ++, mild effect; +++, moderate effect; +++++, large effect.

Propofol

Propofol is an intravenous anesthetic with an alkylphenol molecular structure. Like benzodiazepines, it appears to act on the GABA receptor, although not at the same site as the benzodiazepines. The drug is hydrophobic and is prepared as a lipid emulsion, so it rapidly crosses the blood–brain barrier, leading to a rapid onset of sedation. High lipid solubility also allows for rapid redistribution to the peripheral tissues with a duration of action measured in minutes.⁴⁵ With prolonged infusions, duration may be increased slightly, although it is rare for the effect to last beyond 60 minutes from the time of infusion discontinuation. When the infusion is stopped, the peripheral tissue stores redistribute the drug back into the plasma, but usually not to clinically significant levels because of the drug's high lipid affinity. Propofol is ultimately metabolized predominantly in the liver. It has an elimination half-life of 4 to 7 hours and no active metabolites.

Propofol has predictable pharmacodynamic effects, acting as a hypnotic agent causing a dose-dependent depression of responsiveness and awareness. It is also a potent anxiolytic and amnestic agent.⁴⁵ Indeed, at high infusion rates, propofol is commonly used for general anesthesia. It appears to have effective anticonvulsant properties.⁴⁶ Propofol has no detectable analgesic activity and is not recommended as a sole agent for the initial management of mechanically ventilated patients, because pain control is important in some of these patients. The drug causes ventilatory depression and even apnea in some patients. Because apnea with propofol is

unpredictable and not always dose dependent, it should not be used without readiness to secure the airway. The respiratory pattern with propofol is a decrease in tidal volume and a slight increase in respiratory rate. Propofol can cause significant decreases in blood pressure, especially in hypovolemic patients. Hypotension is mainly caused by preload reduction from dilation of venous capacitance vessels. There is mild myocardial depression that also contributes to hypotension.⁴⁷ The hemodynamic effect is generally more pronounced than with the benzodiazepines and it should be given cautiously to patients with cardiac disease. Hyperlipidemia is a unique side effect of propofol.⁴⁸

As noted above, the drug is delivered in an intralipid carrier, and propofol 1% solution has 1.1 kcal/mL.⁴⁹ Therefore, parenteral lipid feedings must be adjusted to account for the calories administered with this drug. Triglyceride levels should be checked frequently and the drug stopped if hypertriglyceridemia is noted. Strict aseptic technique and frequent changing of infusion tubing is essential to prevent iatrogenic transmission of bacteria and fungi because propofol can support their growth.⁵⁰ A “propofol infusion syndrome,” manifesting as dysrhythmias, heart failure, metabolic acidosis, hyperkalemia, and rhabdomyolysis, is a rare complication. It appears to be more likely to occur when high doses and/or concentrations of propofol are used.⁵¹ An unpublished randomized, controlled trial of propofol in 327 pediatric patients reviewed by the Food and Drug Administration described a concentration-dependent increase in 28-day mortality in propofol-treated patients versus patients treated with other sedatives (4% mortality with sedatives other than propofol, 8% mortality with 1% propofol, and 11% mortality with 2% propofol).⁵² Propofol has received media attention because of problems with national shortages⁵³ and reports of use of this agent as a drug of abuse.^{54,55}

Fospropofol

Fospropofol is a prodrug of propofol that has been introduced as a potential alternative agent for ICU sedation. The Food and Drug Administration approved the drug in 2008. It is metabolized in vivo to the active drug propofol, but the parent drug is water soluble with a much smaller volume of distribution than propofol.⁵⁶ The onset of action is on the order of minutes, just slightly longer than propofol. The potential implications of this characteristic include a lower propensity for accumulating in adipose stores during prolonged infusions, although thus far it has only been studied in Phase III clinical trials for colonoscopy, bronchoscopy,⁵⁷ or minor surgical procedures.⁵⁸ Contamination of fospropofol is less of a concern than propofol because of its water-soluble nature. Further study is needed to evaluate its utility in the ICU.

Butyrophenones

Butyrophenones such as haloperidol may be used for sedation of mechanically ventilated patients. These drugs induce a state of tranquility, and patients often behave with a detached affect. Butyrophenones appear to antagonize dopamine, especially in the basal ganglia, although their exact site of action is not known.

The onset time of intravenous haloperidol is 2 to 5 minutes and the half-life is 2 hours. Starting doses of 1 to 10 mg are typically used with titration depending on the desired end point. The drug is metabolized in the liver.

These drugs are typically reserved for patients who are acutely agitated and hyperactive. A state of cataleptic immobility is occasionally seen. Haloperidol provides no amnesia, no effect on seizure activity, and minimal analgesia. Butyrophenones such as haloperidol are the drugs of choice for agitated delirium. A retrospective study reported an association between haloperidol in ventilated patients and a reduction in mortality,⁵⁹ but there are no confirmatory prospective trials. Currently, there is no convincing evidence for the use of antipsychotics in the treatment of ICU delirium. A small pilot study comparing haloperidol, ziprasidone, and placebo given to delirious patients in a preemptive manner found no differences in outcomes, although this study may have been underpowered to detect a difference.³⁴ A randomized trial comparing haloperidol to olanzapine showed similar reductions in delirium, but fewer extrapyramidal side effects with olanzapine.⁶⁰

Haloperidol causes no significant respiratory depression.⁶¹ This is an attractive feature of haloperidol, because most sedative or analgesic drugs cause respiratory depression.

Haloperidol is known to prolong the QT interval in some patients; it has been reported to result in torsades de pointes,⁶² although this problem is rare. The drug also mildly antagonizes the α_1 receptor and may decrease the neurotransmitter function of dopamine, resulting in mild hypotension.

Extrapyramidal effects are occasionally seen with these drugs, but are much less common with intravenous than with oral butyrophenones. When these complications occur, treatment with [diphenhydramine](#) or benztropine may be necessary. The neuroleptic malignant syndrome is another extremely rare problem, thought to result from central dopaminergic blockade leading to extrapyramidal side effects, muscle rigidity, and excess heat generation. It is a life-threatening complication, manifested by “lead pipe” muscle rigidity, fever, and mental status changes. Although data are limited, dantrolene, bromocriptine, and benzodiazepines are listed as possible treatment options.⁶³

Dexmedetomidine

Dexmedetomidine is a selective α_2 agonist with both sedative and analgesic properties.⁶⁴ Patients receiving this drug are sedated when undisturbed, but they may arouse easily with minimal stimulation, particularly if they are not receiving opiates or other sedatives concurrently and do not have encephalopathy related to their critical illness. Dexmedetomidine is unique in that some patients transition from sedated to alert states quickly and easily without needing to discontinue the drug infusion, which permits frequent neurologic examinations. The drug is approved for short-term use (<24 hours) in patients initially receiving mechanical ventilation.

Dexmedetomidine is analgesic sparing in postoperative patients.⁶⁵ Because it causes no respiratory depression, it can be continued after discontinuation of mechanical ventilation and extubation. Because it does not produce respiratory depression, it is not a desirable agent when this property is beneficial, such as the initial management of a ventilated patient in whom permissive hypercapnia is sought.

Side effects include bradycardia (on rare occasions of a severe nature⁶⁶) and hypotension,⁶⁷ especially with hypovolemia or high endogenous sympathetic tone. Vasoconstriction and hypertension with increasing doses of dexmedetomidine also have been described.⁶⁸ Early studies with dexmedetomidine evaluated postoperative ICU patients⁶⁹; however, more recent studies have targeted other types of patients receiving dexmedetomidine for longer periods of time. Pandharipande et al reported that dexmedetomidine resulted in more days alive without delirium or coma and more time at the targeted level of sedation than lorazepam.⁷⁰ Riker et al studied dexmedetomidine for longer-term ICU infusions and found that, compared to midazolam, it resulted in less ICU delirium and fewer days on the ventilator.⁷¹ [Table 50-3](#) summarizes the pharmacologic properties of the other commonly used sedative agents: propofol, haloperidol, and dexmedetomidine.

Table 50-3: Properties of Other Sedative Agents

	Propofol	Fospropofol	Haloperidol	Dexmedetomidine
Typical starting dose	1 to 2 mg/kg	2 to 6 mg/kg	0.5 to 1 mg	0.5 to 1.0 mcg/kg over 10 minutes; 0.2 to 0.7 mcg/kg/hour infusion
Onset	0.5 to 1 minutes	4 to 12 minutes	2 to 5 minutes	5 to 10 minutes
Duration after single dose	2 to 8 minutes	2 to 8 minutes	2 hours	30 to 60 minutes
Metabolism	Hepatic, renal	Hepatic, renal	Hepatic	Hepatic
Elimination	Renal	Renal	Renal	Renal
Anxiolysis	++++	++++	+++	+++
Analgesia	No effect	No effect	No effect	++
Hypnosis	++++	++++	++	+++
Amnesia	++++	++++	No effect	+
Seizure threshold	Increases	Increases	No effect	No effect
Reducing dyspnea	+	+	No effect	No effect
Cardiovascular effect	Venodilation, arteriolar dilation, myocardial depression	Venodilation, arteriolar dilation, myocardial depression	Venodilation, arteriolar dilation	Venodilation, arteriolar dilation, bradycardia, occasional hypertension
Respiratory effect	Hypoventilation	Hypoventilation	No effect	No effect
Common side effects	Increased triglycerides	Increased triglycerides	Neuroleptic malignant syndrome (rare), extrapyramidal effects (rare)	Hypotension, bradycardia

+, minimal effect; ++, mild effect; +++, moderate effect; +++++, large effect.

Ketamine

Ketamine has a molecular structure similar to phencyclidine. Patients experience a state of mind in which perception is separated from sensation, a “detached from surroundings” or so-called dissociative state. While appearing unaware of their surroundings, patients keep their eyes open and maintain a protective cough reflex. They may behave with coordinated movements that appear to be without purpose. The drug has profound analgesic properties without producing respiratory depression. The circulatory effects of hypertension

and tachycardia, typically seen, reflect increased activity of the sympathetic nervous system. Some, but not all patients, experience amnesia after receiving ketamine. A recent study found ketamine to be an effective alternative to etomidate to facilitate intubation in the ICU. There was a significantly lower incidence of adrenal insufficiency with ketamine.⁷² The side effects of emergence delirium and severe hallucinations limit ketamine as a mainstream agent for sedation and analgesia in mechanically ventilated patients. Indeed, this phencyclidine derivative is popular as an illicit drug of abuse.

Barbiturates

Barbiturates such as thiopental and pentobarbital are potent agents that cause amnesia and unconsciousness. They have no role as sedatives in ICU patients because of a propensity to cause hemodynamic instability and prolonged accumulation in peripheral tissues secondary to lipid solubility. Thiopental is sometimes used to induce anesthesia to facilitate endotracheal intubation. These drugs may be used to induce a pharmacologic coma in patients with severe brain injury.

Inhalational Anesthetics

Inhalational anesthetics are used often in the operating room to maintain general anesthesia in mechanically ventilated patients. These agents have analgesic, amnestic, and hypnotic sedating properties. These exhaled gases must be effectively scavenged because they are not metabolized to any significant degree. Delivery and scavenging are technically challenging problems, which has limited greatly the use of these agents outside of the operating room. While there are devices that can be attached to ventilator circuitry to recycle inhalational anesthetics,^{73,74} these agents are largely restricted to the research setting and more data are needed before their widespread use in the ICU.

Strategies for Delivery of Sedatives in Critically Ill Mechanically Ventilated Patients

Because no single drug can achieve all of the indications for sedation and analgesia in the ICU, a combination of drugs, each titrated to specific end points, may be more effective. This strategy can allow lower doses of individual drugs and reduce drug accumulation. Both continuous intravenous infusion and intermittent bolus techniques have been advocated. Intermittent bolus strategies may lead to fluctuations in level of sedation and increase demands on nursing time, potentially distracting attention away from other patient care issues. The perceived benefits of continuous sedative infusions include a more consistent level of sedation with better patient comfort, though convenience is likely the greatest reason for its popularity.

Ideally, strategies for sedation and analgesia in critically ill patients should adhere to pharmacokinetic and pharmacodynamic principles. Unfortunately, ICU patients frequently exhibit unpredictable alterations in pharmacology so that precise guidelines for drug administration are not possible. For instance, when short-acting benzodiazepines such as midazolam and lorazepam are administered in the ICU, these drugs accumulate in tissue stores with consequent prolonged clinical effect. Other circumstances that confound prediction of pharmacologic behavior of sedatives and analgesics include altered hepatic and/or renal function, polypharmacy in the ICU with complex drug-drug interactions, altered protein binding, and circulatory instability. The multicompartmental pharmacokinetics typical in critically ill patients defy simple bedside pharmacokinetic profiling.⁷⁵ As such, titration of sedatives and analgesics against discernible clinical end points, while imprecise, is the most commonly utilized tool. Further confounding administration of sedatives in the ICU is the dramatic difference between extremes of sedation. Because oversedated patients are easier to manage than undersedated patients, clinicians may be heavy handed when sedating agitated patients.

The occasional ICU patient may require extraordinarily high doses of sedatives to achieve tranquility; such doses may be much greater than those quoted in the literature and recommended by drug manufacturers. Indeed, as discussed below in “[Neuromuscular Blocking Agents](#),” occasional patients may even require pharmacologic paralysis to achieve synchrony with mechanical ventilation.⁷⁶

As sicker patients demonstrate improved outcomes in the ICU,^{76–80} aggressive levels of sedation and analgesia are sometimes necessary. This is particularly likely for patients managed with unconventional ventilator strategies (e.g., permissive hypercapnia, low tidal volumes, prone positioning, and pressure-controlled ventilation), because these strategies may be inherently distressing to many patients. For selected patients, deep sedation may be the only practical option.

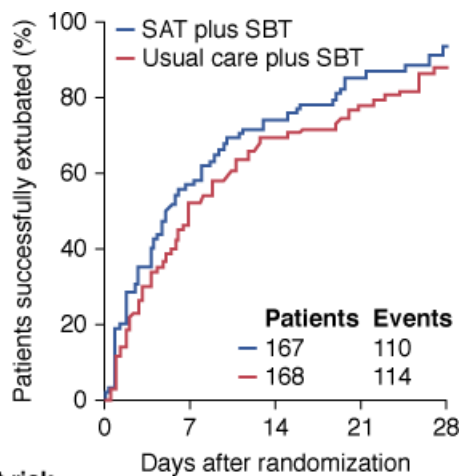
The use of deep sedation, however, carries a high price, because the neurologic examination is severely limited. Ideally, a head-to-toe daily assessment for the presence of organ failures should be routine for every critically ill patient. This is particularly so during resuscitative phases of care, when assessing the adequacy of end-organ perfusion and function is vital. The mental status examination is an important gauge of brain perfusion. Because brain injury is a devastating complication of critical illness,⁸¹ acute cerebral dysfunction must be detected quickly and corrected if possible before permanent injury takes place. The veil of sedation severely handicaps clinicians' ability to serially follow a patient's neurologic condition.

A protocol-driven approach to sedation has been shown to alleviate many of the problems mentioned above.⁸² Such protocols assure adequate analgesia and sedation using frequent assessments of patient needs with goal-directed titration of analgesics and sedatives. Alternatively, a routine protocol of daily interruption of continuous sedative infusions can reduce many of the complications of sedation in the ICU setting, including duration of mechanical ventilation and ICU length of stay.¹⁷ Such a strategy allows patients to spend some of their ICU time awake and interactive. This can allow earlier mobilization of intubated patients leading to improved independence of function and reduced ICU delirium,⁸³ as well as reduced drug administration and reduced diagnostic neurologic evaluations. Such protocol-driven sedation strategies allow a focused downward titration of sedative infusion rates over time, streamlining administration of these drugs and minimizing the tendency for accumulation.

Initially, the thought of decreasing or stopping sedatives in a critically ill patient who has been agitated may be unsettling. As such, clinicians may aggressively sedate patients early in their ICU course, and maintain the same level of deep sedation indefinitely. A daily holiday from sedatives can eliminate the tendency to "lock in" to a high sedative infusion rate. When sedative infusions are decreased or stopped, tissue stores can redistribute drug back into the circulation. Sometimes interruption of sedative infusions can lead to abrupt awakening and agitation. This must be anticipated by the ICU team to avoid complications such as patient self-extubation; if excessive agitation is noted, sedatives should be restarted. Although the attempt at waking and communication may fail on a given day, this does not portend inevitable failure on all subsequent days. When awakening patients from sedation, reaching the brink of consciousness, without precipitating excessive agitation may be ideal for some. Once objective signs of consciousness are demonstrated, restarting sedatives at half the previous dose and *as needed* is reasonable. Adjustments from this starting point can be individualized to patient needs.

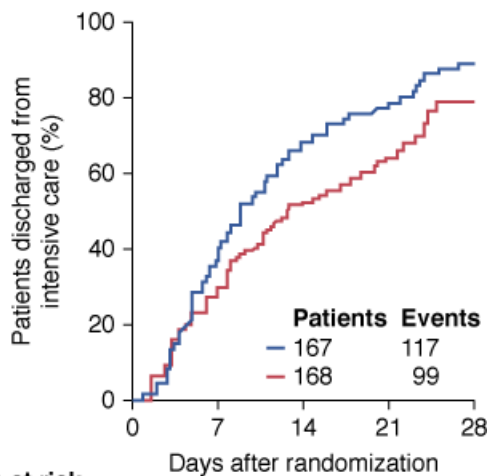
It is clear that sedatives may impact the duration of mechanical ventilation.^{17,84} Pairing sedation strategies, such as daily sedative interruption with spontaneous breathing trials, results in shortened ventilator, ICU, and hospital days⁸⁵ (Fig. 50-1).

Figure 50-1.



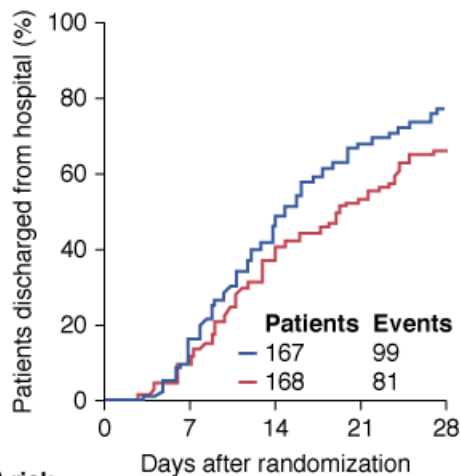
Patients at risk

	0	7	14	21	28
SAT plus SBT	167	57	24	9	3
Usual care plus SBT	168	68	30	18	8

A

Patients at risk

	0	7	14	21	28
SAT plus SBT	167	89	35	20	10
Usual care plus SBT	165	102	52	33	18

B

Patients at risk

	0	7	14	21	28
SAT plus SBT	167	126	64	34	24
Usual care plus SBT	165	130	72	47	30

C

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Probability of successful extubation (A), discharge from intensive care (B), and hospital discharge (C) during the first 28 days after randomization. SAT, spontaneous awakening trial; SBT, spontaneous breathing trial. (Used, with permission, from Girard et al.⁸⁵)

Literature evaluating long-term consequences of recovery from respiratory failure and sedation suggests that post-ICU depression is common in patients who require mechanical ventilation during critical illness.⁸⁶ Posttraumatic stress disorder following recovery has been reported as well.^{29,31} Some data suggest that lack of awareness related to sedation and/or underlying illness is associated with development of this disorder, and that preservation of awareness during mechanical ventilation may reduce this problem.^{29,31,87}

Neuromuscular Blocking Agents

Indications

The use of neuromuscular blocking agents (NMBAs) during mechanical ventilation has decreased considerably in the last two decades. These agents remain a common tool to facilitate endotracheal intubation and assure immobility during surgical procedures. The need for these drugs beyond their use to facilitate procedures is relatively rare in the ICU. Complications related to the use of NMBAs during critical illness have relegated these agents to use as a last resort, usually only for extreme derangements in cardiopulmonary physiology.

Indications for neuromuscular blockade include facilitation of mechanical ventilation in the rare patient who continues to have extreme *ventilator dyssynchrony* despite aggressive administration of sedatives and analgesics. Occasionally, a patient remains so dyssynchronous with the ventilator that effective ventilation is not possible and *severe derangements in gas exchange* ensue, leading to a need for neuromuscular blockade.

Papazian et al conducted a randomized trial in patients with severe acute respiratory distress syndrome (partial pressure of arterial oxygen-to-fractional inspired oxygen concentration ratio $[PaO_2/FiO_2] < 150$), in which patients received cisatracurium versus placebo in a double-blinded manner. Those receiving cisatracurium had a better adjusted 90-day mortality (hazard ratio [HR] 0.68 [0.48 to 0.98]; $P = 0.04$).⁷⁶ The mechanism(s) behind the survival benefit in this study is not known, but may relate to patients being more synchronous with the ventilator and hence avoiding alveolar overdistension or opening-closing cycles that lead to ventilator-associated lung injury.

Patients with *tetanus* may require neuromuscular blockade because of chest wall rigidity, which may prohibit effective chest wall excursion and ventilation. As mentioned above in the discussion of opiate toxicities, the rare occurrence of *skeletal muscle rigidity with high-dose synthetic opiates* may require neuromuscular blockade to permit ventilation. This problem is rare and muscle rigidity short-lived, so that neuromuscular blockade is necessary for only a short time. Lastly, circulation in patients with *severe hemodynamic instability* may become more compromised when respiratory distress demands even greater blood flow to the respiratory muscles from a circulation that cannot accommodate such demands. It is mandatory that patients given NMBAs receive drugs to assure amnesia while they are pharmacologically paralyzed.

Assessing Level of Blockade

Acetylcholine is released from synaptic vesicles at the terminal end of the motor nerve and binds to the postsynaptic end plate, propagating an electrical signal and leading to muscle contraction. Pharmacologic NMBAs bind to the acetylcholine receptor at the terminal end of the motor nerve. These agents can activate the acetylcholine receptor (depolarizing agents) or competitively inhibit the receptor without activating it (nondepolarizing agents). The depth of neuromuscular blockade from nondepolarizing agents is most accurately monitored with the use of a peripheral nerve stimulator. The use of a peripheral nerve stimulator in the ICU has been shown to reduce the amount of neuromuscular blocking drug used and shorten recovery of neuromuscular function and spontaneous ventilation;⁸⁸ however, with cisatracurium as the preferred ICU neuromuscular blocking agent (see below the section of Agents–Non depolarizing neuromuscular blocking agents), such monitoring may be unnecessary given the reliable clearance of this drug.⁸⁹

Selection of Agent

Depolarizing Neuromuscular Blocking Agents

Succinylcholine

Succinylcholine is the only available depolarizing NMBA. It has the most rapid and reliable onset of neuromuscular blockade, with paralysis sufficient to permit endotracheal intubation within 60 seconds of dosing. As such, succinylcholine is used only to facilitate endotracheal intubation but is not indicated for ongoing neuromuscular blockade in critically ill patients.

Nondepolarizing Neuromuscular Blocking Agents

A number of nondepolarizing NMBAs are currently available. The pharmacology of the drugs commonly used during mechanical ventilation in the ICU is discussed below.

Pancuronium

Pancuronium is an older NMBA used during general anesthesia. It has a long half-life, with a duration of action between 60 and 90 minutes after a single intravenous bolus dose of 0.1 mg/kg. Its vagolytic side effect (typically results in a heart rate increase of approximately 10 beats/min), active metabolite (3-hydroxypancuronium), and reliance on renal clearance limit its attractiveness in critically ill patients.

Vecuronium

After the usual loading dose of 0.1 mg/kg, vecuronium typically lasts 30 minutes. Half of the drug is excreted in bile, so prolonged action may be seen in patients with liver dysfunction. In addition, one-third of the drug is excreted in the kidneys, so that accumulation in the setting of renal insufficiency may be seen. The active metabolite 3-desacetylvecuronium may lead to prolongation of effect with repeated dosing, particularly in patients with renal failure.⁹⁰

Rocuronium

Unlike the other aminosteroidal nondepolarizing NMBAs, rocuronium has a rapid onset of action, typically within 60 to 90 seconds. It may be used to facilitate endotracheal intubation, as a substitute for succinylcholine, when the latter is contraindicated (e.g., burns, muscle tissue injury, and upper motor neuron lesions). The usual bolus dose is 0.6 to 1.0 mg/kg, with a duration of effect of 30 to 45 minutes, similar to vecuronium. The metabolite, 17-desacetylrocuronium, has minimal neuromuscular blocking activity.

Atracurium

Atracurium is a benzylisoquinolinium compound with a duration of action between 20 and 45 minutes. The initial loading dose is 0.4 to 0.5 mg/kg. The drug is usually given by continuous infusion in critically ill, mechanically ventilated patients at a dose of 10 to 20 mcg/kg/min. Atracurium is inactivated in plasma by ester hydrolysis and Hofmann elimination so that renal or hepatic dysfunction do not impact its duration of blockade. Atracurium may cause histamine release, and its breakdown product, laudanosine, has been associated with central nervous system excitation and seizures. Currently, it is rarely used in the ICU.

Cisatracurium

Cisatracurium is an isomer of atracurium and it has a similar pharmacologic profile to atracurium. The initial loading dose is 0.1 to 0.2 mg/kg and the duration of action is approximately 25 minutes. Like atracurium, this drug is inactivated in plasma by ester hydrolysis and Hofmann elimination. Cisatracurium does not cause histamine release. Because of its short half-life, it requires administration by continuous infusion. The usual dose is 2.5 to 3 mcg/kg/min. This drug is currently the most frequently used for neuromuscular blockade in mechanically ventilated ICU patients. Apart from succinylcholine and rocuronium use to facilitate endotracheal intubation, there seems little reason to ever use a neuromuscular blocking agent other than cisatracurium in the ICU.

Toxicity

In normal individuals, depolarization of skeletal muscle beds after administration of depolarizing NMBAs such as succinylcholine leads to release of intracellular potassium, typically resulting in an increase in serum potassium of approximately 0.5 mEq/L. Denervation of skeletal muscle from tissue injury, such as with burns or upper motor neuron lesions, may result in more dramatic rises in serum potassium with administration of succinylcholine.

The major toxicity with nondepolarizing NMBAs is prolonged weakness, which occurs by two separate mechanisms. The accumulation of a NMBA parent drug or its metabolites is seen with some agents (e.g., pancuronium and vecuronium), especially in patients with renal and/or hepatic dysfunction.⁹¹ The second cause of weakness associated with NMBAs is known as *acute quadriplegic myopathy syndrome*. Patients with this syndrome manifest acute paresis, myonecrosis with increased creatine phosphokinase concentration, and abnormal electromyography. Electromyographic findings are consistent with denervation of skeletal muscle, which may progress to muscle atrophy and even muscle necrosis.

Complications of NMBAs have reduced, but not eliminated, their use in the ICU.⁹² This is particularly noteworthy when corticosteroids are used in conjunction with NMBAs. Several studies suggest this combination is associated with a significant incidence of myopathy.⁹³ [Table 50-4](#) summarizes the pharmacologic properties of commonly used NMBAs.

Table 50-4: Properties of Commonly Used Neuromuscular Blocking Agents

	Succinylcholine	Pancuronium	Vecuronium	Rocuronium	Atracurium	Cisatracurium
Type	Depolarizing	Nondepolarizing	Nondepolarizing	Nondepolarizing	Nondepolarizing	Nondepolarizing
Typical loading/intubation dose	1.0 to 1.5 mg/kg	0.1 mg/kg	0.1 mg/kg	0.6 to 1.0 mg/kg	0.5 mg/kg	0.1 to 0.2 mg/kg
Continuous infusion? (dose)	No	No	No	No	Yes (10 to 20 mcg/kg/min)	Yes (2 to 3 mcg/kg/min)
Onset	60 seconds	3 to 5 minutes	3 to 5 minutes	1 to 2 minutes	3 to 5 minutes	3 to 5 minutes
Duration after loading dose	7 to 9 minutes	60 to 90 minutes	30 to 40 minutes	30 to 45 minutes	20 to 45 minutes	25 minutes
Metabolism	Plasma cholinesterase	Hepatic	Hepatic/biliary	Hepatic	Ester hydrolysis/Hoffman elimination	Ester hydrolysis/Hoffman elimination
Elimination		Renal	Renal/biliary	Biliary clearance of parent molecule from plasma		
Active metabolite?	No	Yes	Yes	Yes (minimal)	No	No
Cardiovascular effect	Supraventricular and ventricular rhythm disturbances with hyperkalemia	Vagolytic—tachycardia	None	None	Histamine release	No histamine release
Common side effects	Hyperkalemia	None	None	None	Histamine release	None

Important Unknowns and the Future

Currently available drugs for pain control, sedation, and neuromuscular blockade in mechanically ventilated patients all have important limitations. Given the availability of agents with fewer propensities to accumulate and impair neurologic evaluations or liberation from mechanical ventilation, the use of benzodiazepines in the ICU is falling out of favor. Novel approaches, such as inhalational agents and fospropofol, will require more research before they move into the mainstream.

Protocol-driven management of sedation and pain control can clearly improve patient outcomes, regardless of the agent administered. More experience with protocols and better understanding of potential limitations is needed to continue to optimize patient outcomes, both short-term and long-term. As such, further expansion of protocol-driven administration of sedation and analgesia should be evaluated, particularly in different ICU environments, such as general surgery, trauma, or neurologically injured patients.

Use of sedation scales and titration of drugs based upon goals described by these objective scales should allow optimization of sedation and analgesia. The impact of sedation and analgesia on delirium during critical illness must be further studied, given the magnitude of this problem in terms of prevalence and associated morbidity. The impact of neuromuscular blockade, sedation, and pain control on recovery following respiratory failure is poorly understood. Patients who require mechanical ventilation during critical illness suffer psychological and physical complications, and recovery may be slow and incomplete.^{94–96} With recognition of psychological maladjustment, a few strategies achieving improved outcomes have been reported,^{31,87,97} but more studies in this area are needed. Physical debilitation after respiratory failure and critical illness is well described, with most studies focusing on recovery from acute respiratory distress syndrome. Many of these patients suffer from critical illness-related polyneuropathy and myopathy.^{98,99} Sedation strategies can allow patients to be more interactive in the ICU environment; this can encourage the opportunity for very early mobilization in the ICU^{83,100,101} and reductions in functional impairment and deconditioning.

Summary and Conclusion

Pain control, sedation, and neuromuscular blockade are important components of the treatment of mechanically ventilated patients. Directing treatment to specific and individualized goals will assure that patient needs are met. All currently available agents for use in ventilated patients have limitations and complications related to their use. Rather than seeking an ideal drug, strategies of drug administration that focus attention on principles of sedative pharmacology in critical illness should be utilized. When these drugs are given to individual patients, recognition of specific goals will allow rational administration strategies to be implemented, which should lead to improvement in short- and long-term outcomes.

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